PPG-DRIVEN BP ESTIMATION

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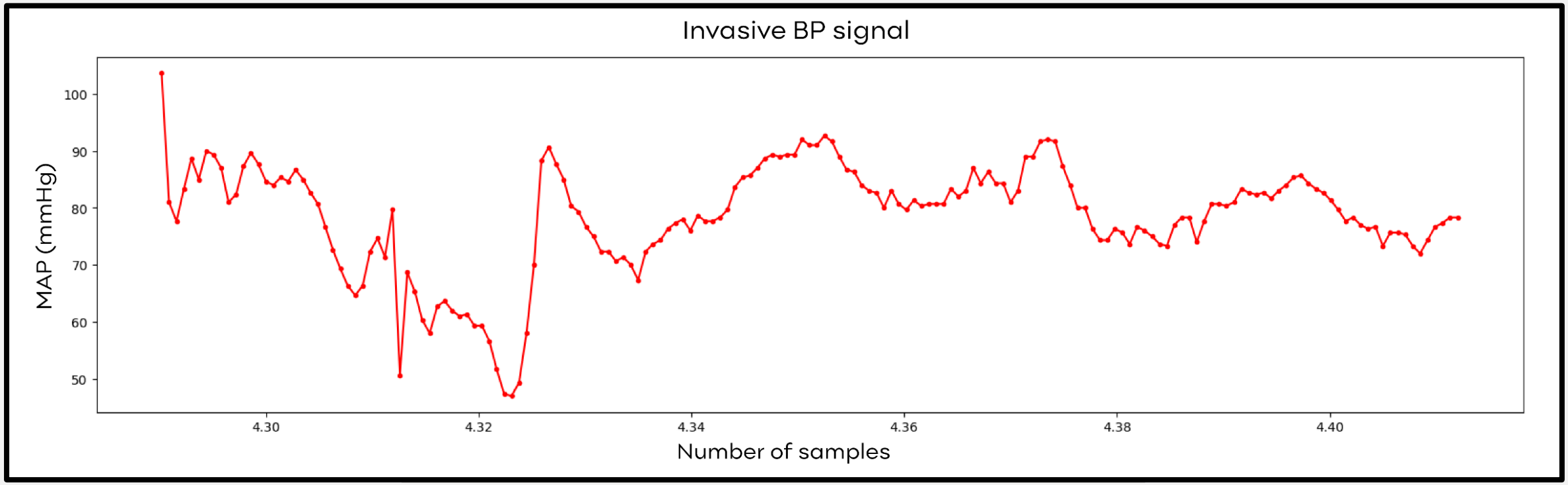
BLOOD PRESSURE

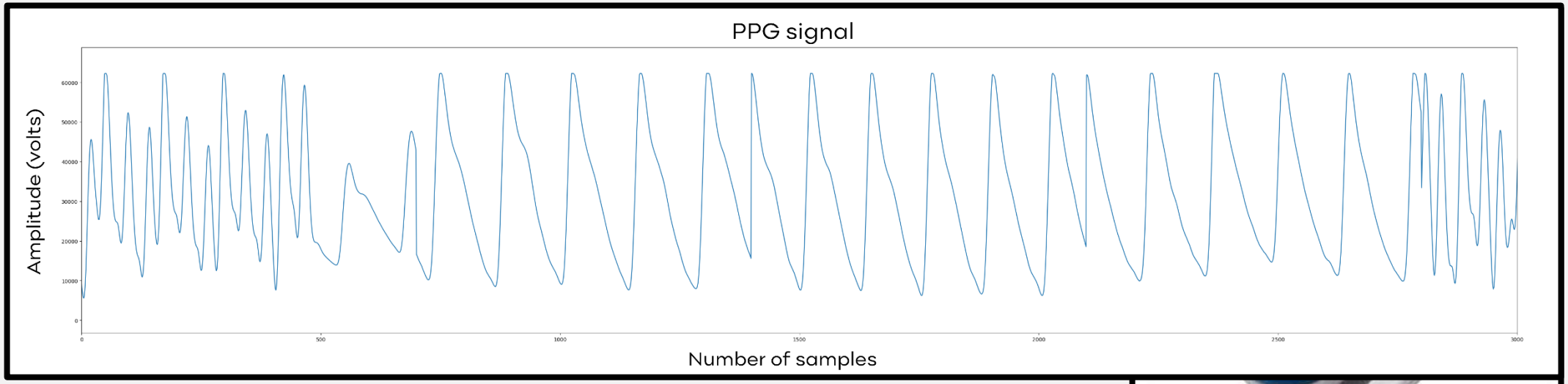
Blood pressure (BP) monitoring is essential for detecting complications during surgeries.

Measurements of BP currently



PPG

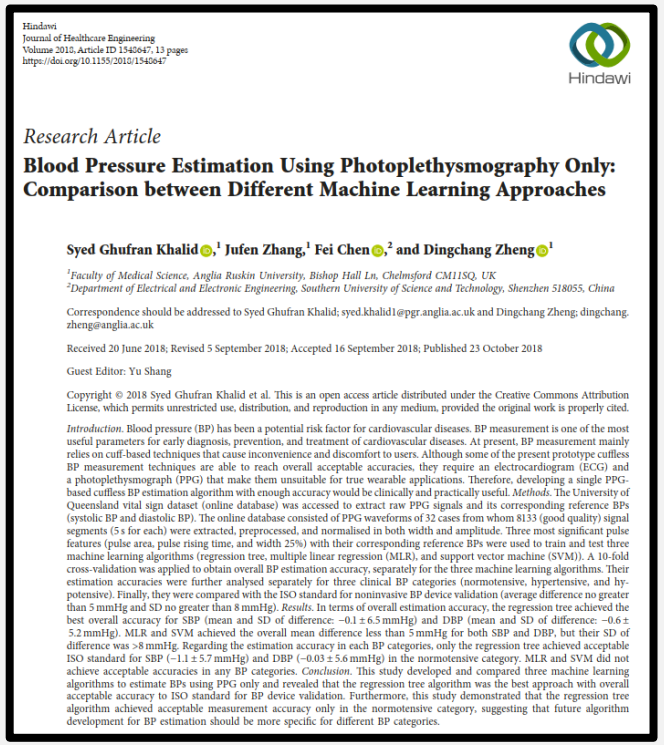
- A non-invasive sensor used to assess blood flow in the skin using infrared
- Potential signal for extracting and evaluating the BP value continuously



PROJECT OVERVIEW

- **Project goal:** Investigate methods for continuous non-invasively detecting BP using a single PPG sensor to be used during surgery under anesthesia
- Cooperation with ‘Nervio’
- Finding a relevant article

NERVIO
Neurophysiological AI



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Research Article

Blood Pressure Estimation Using Photoplethysmography Only: Comparison between Different Machine Learning Approaches

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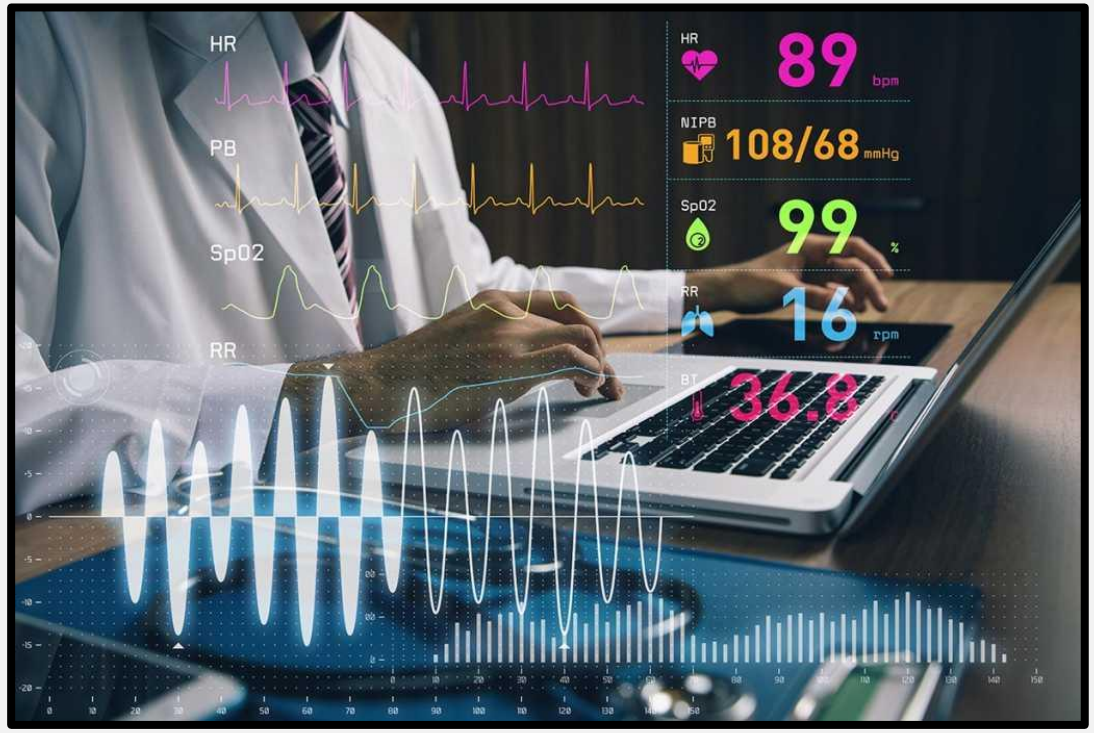
Introduction. Blood pressure (BP) has been a potential risk factor for cardiovascular diseases. BP measurement is one of the most useful parameters for early diagnosis, prevention, and treatment of cardiovascular diseases. At present, BP measurement mainly relies on cuff-based techniques that cause inconvenience and discomfort to users. Although some of the present prototype cuffless BP measurement techniques are able to reach overall acceptable accuracies, they require an electrocardiogram (ECG) and a photoplethysmograph (PPG) that make them unsuitable for true wearable applications. Therefore, developing a single PPG-based cuffless BP estimation algorithm with enough accuracy would be clinically and practically useful. *Methods*. The University of Queensland vital sign dataset (online database) was accessed to extract raw PPG signals and its corresponding reference BPs (systolic BP and diastolic BP). The online database consisted of PPG waveforms of 32 cases from whom 8133 (good quality) signal segments (5 s for each) were extracted, preprocessed, and normalised in both width and amplitude. Three most significant pulse features (pulse area, pulse rising time, and width 25%) with their corresponding reference BPs were used to train and test three machine learning algorithms (regression tree, multiple linear regression (MLR), and support vector machine (SVM)). A 10-fold cross-validation was applied to obtain overall BP estimation accuracy, separately for the three machine learning algorithms. Their estimation accuracies were further analysed separately for three clinical BP categories (normotensive, hypertensive, and hypotensive). Finally, they were compared with the ISO standard for noninvasive BP device validation (average difference no greater than 5 mmHg and SD no greater than 8 mmHg). *Results*. In terms of overall estimation accuracy, the regression tree achieved the best overall accuracy for SBP (mean and SD of difference: -0.1 ± 6.5 mmHg) and DBP (mean and SD of difference: -0.6 ± 5.2 mmHg). MLR and SVM achieved the overall mean difference less than 5 mmHg for both SBP and DBP, but their SD of difference was >8 mmHg. Regarding the estimation accuracy in each BP categories, only the regression tree achieved acceptable ISO standard for SBP (-1.1 ± 5.7 mmHg) and DBP (-0.03 ± 5.6 mmHg) in the normotensive category. MLR and SVM did not achieve acceptable accuracies in any BP categories. *Conclusion*. This study developed and compared three machine learning algorithms to estimate BPs using PPG only and revealed that the regression tree algorithm was the best approach with overall acceptable accuracy to ISO standard for BP device validation. Furthermore, this study demonstrated that the regression tree algorithm achieved acceptable measurement accuracy only in the normotensive category, suggesting that future algorithm development for BP estimation should be more specific for different BP categories.

DATA

- 67 patients anesthetized
- Signals of PPG sensor (100 Hz)
- Invasive BP (1/60 Hz)- 92%
- Non-invasive (irregular rate)
- Patient’s meta data



ALL PROCESS

Data preparation → Train & test split → Pre-processing → Feature extraction → Predictive model

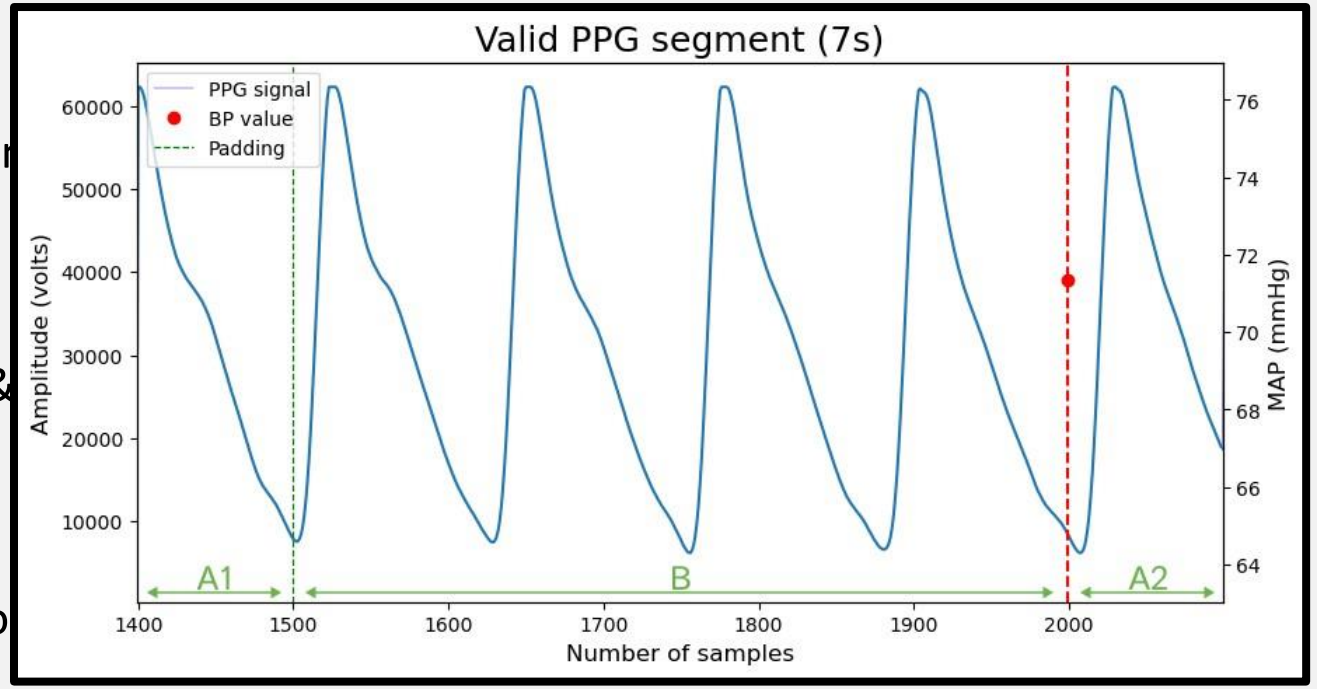
Extraction of **6303** valid PPG segments around a valid BP measurement

80% & 20%

4212 processed and clear

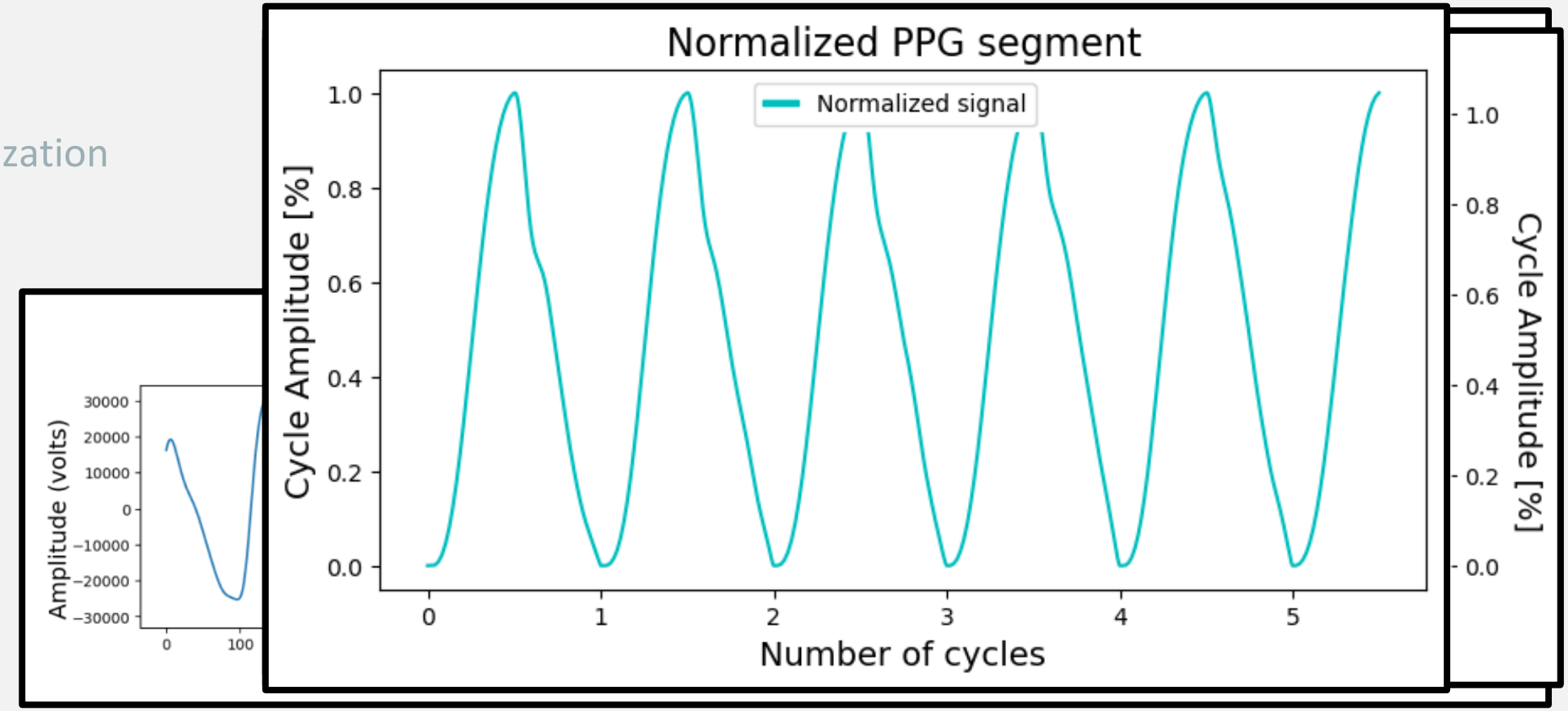
125 waveform features &

Regression & classificatio



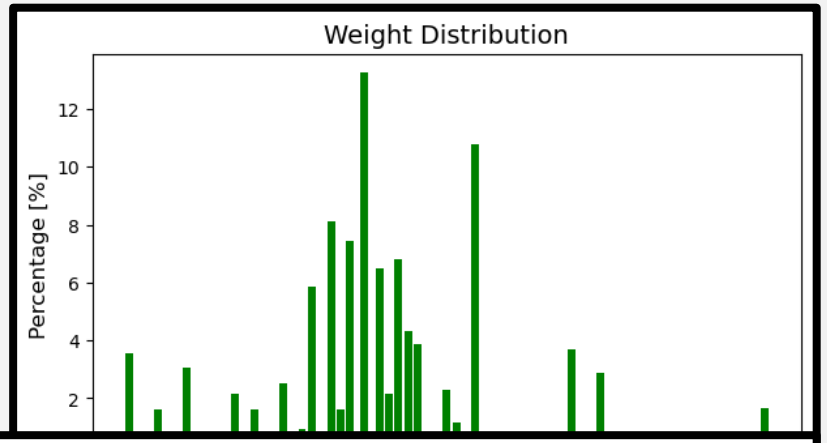
PRE-PROCESSING

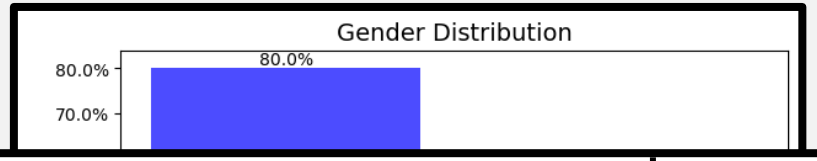
- Noise filtering
- Trend removal
- Finding cycles
- Segments disqualified
- 2-dimensional normalization
- Cut to whole cycles

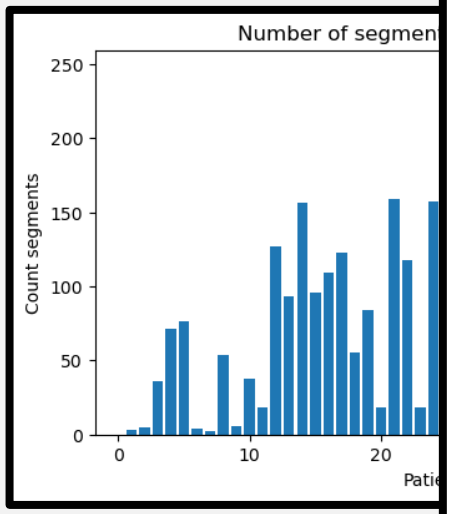


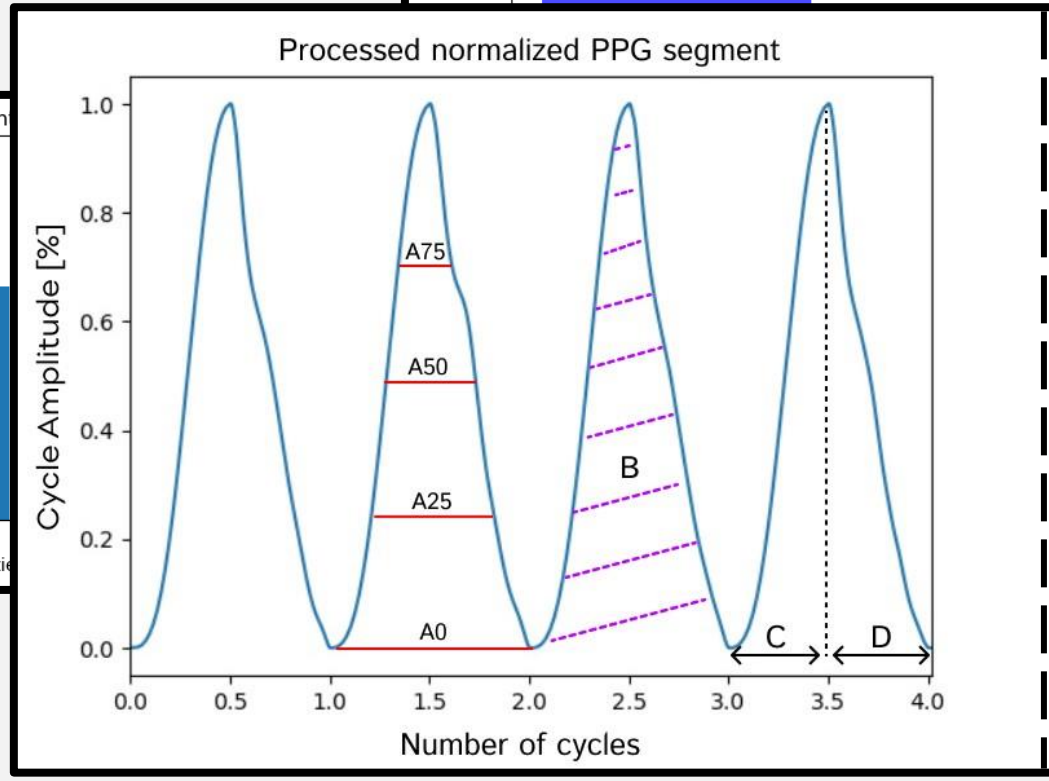
FEATURE EXTRACTION

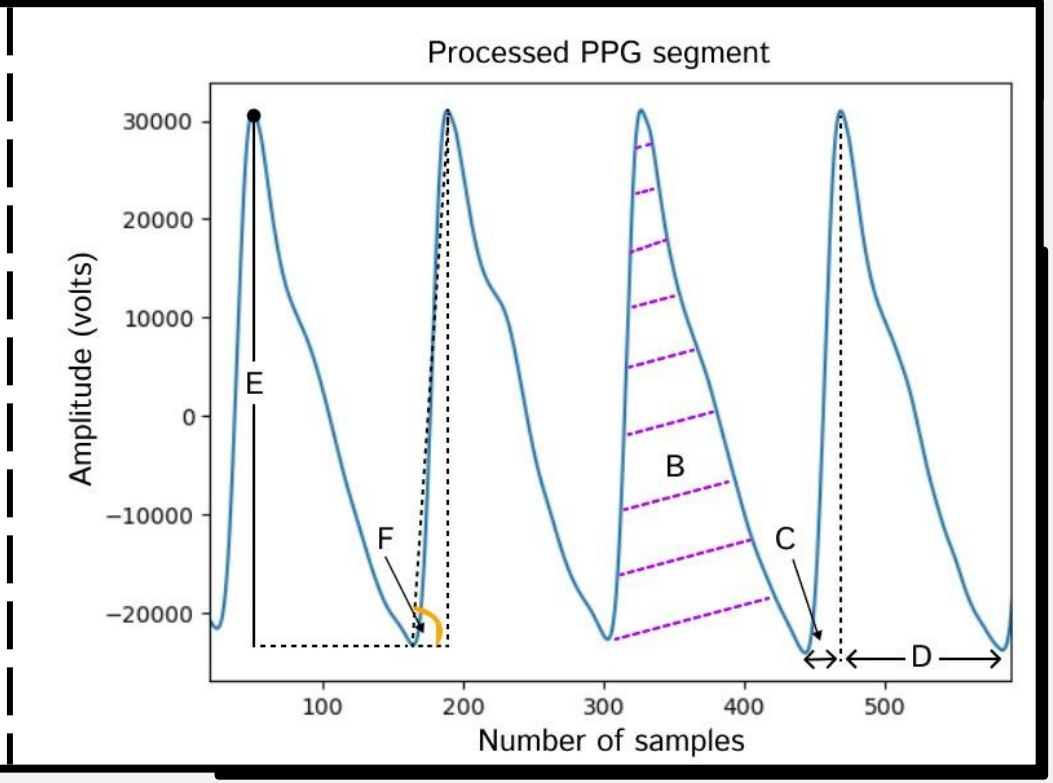
- **Defining 4 labels- DMAP, RMAP, SBP, DBP**
- Defining 125 features
- EDA







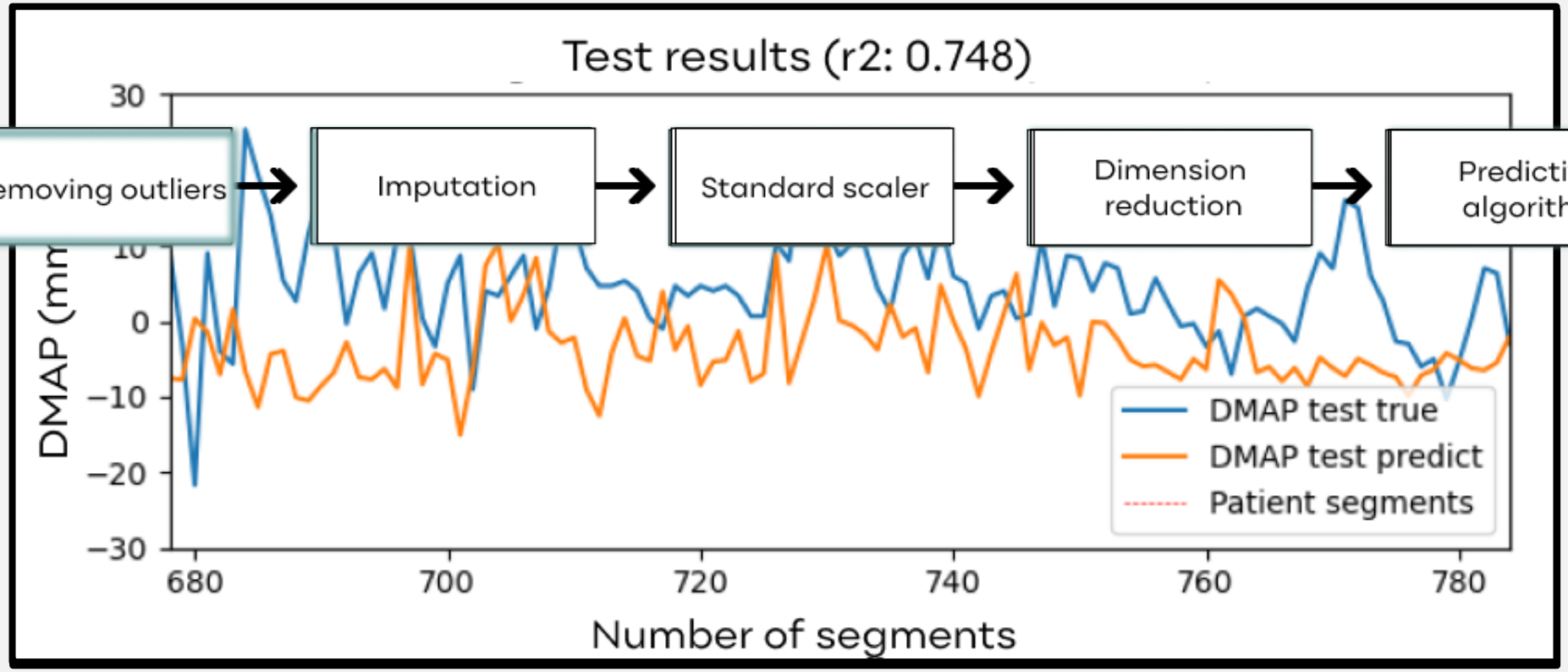




PREDICTIVE MODEL

SVM regression model predicting differential BP change (DMAP).

A close-up of a single patient's results



Removing outliers → Imputation → Standard scaler → Dimension reduction → Predictive algorithm → Adjusting hyperparameters

Removing outliers → Imputation → Adjusting hyperparameters → Dimension reduction → Predictive algorithm → Adjusting hyperparameters

RESULTS ANALYSIS

- BP index was better predicted with our improvements
- Baseline feature serves as calibration
- Model learns from all data
- Regressions outperformed classifications

DISCUSSIONS

- PPG-based BP prediction system shows potential
- Data quality
- AAMI standards for over 85 patients
- Measurement cut dispersion
- More waveform features
- Deep-learning for BP prediction

THANKS FOR LISTENING